

M 292/95

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Total No. of printed pages = 03

Monsoon, 2024

B. Tech. 1st Semester Examination

Biology for Engineers

Course Code: BBI23101T

Full Marks – 50

Time –2 hours

The figure in the margin indicates full marks for the questions.

1) Multiple choice question (each question will carry 1 mark)

10x1=10

A) Who discovered the double helix structure of DNA?

- i) Louis Pasteur
- ii) James Watson and Francis Crick
- iii) Robert Hooke
- iv) Hugo de Vries

B) Which biologist has discovered the word cell?

- i) Robert Hooke
- ii) Robert Brown
- iii) Louis Pasteur
- iv) None of these

C) How many DNA molecules are there in each chromosome?

- i) One
- ii) Two
- iii) Ten
- iv) Infinite

D) Into how many groups are plants divided?

- i) 6
- ii) 7
- iii) 5
- iv) 4

E) Which of the following has great diversity?

- i) Plants
- ii) Animals
- iii) Fossils
- iv) Insects

- F) The study of kinds and diversity of organisms and the evolutionary relationship among them is called _____?
- i) Taxonomy
 - ii) Physiology
 - iii) Biology
 - iv) Chemistry
- G) Which of the following is not a macromolecule?
- i) Protein
 - ii) Carbohydrate
 - iii) Lipid
 - iv) Vitamin
- H) The alternate form of a gene is
- i) Recessive character
 - ii) Alternate type
 - iii) Allele
 - iv) Dominant Character
- I) The genotypic ratio of a monohybrid cross is
- i) 3:1
 - ii) 1:2:1
 - iii) 9:3:3:1
 - iv) 2:1:1
- J) Pea plants were used in Mendel's experiments because
- i) They were cheap
 - ii) They were available easily
 - iii) They had contrasting characters
 - iv) All the above
- 2) A) Why we need to study Biology? What is its significance in studying technology courses? What is Brownian motion? Explain with a suitable diagram. 2+3+3+2=10
- Or
- B) What is the fundamental difference between Science and Technology? Try to explain the same by drawing comparison between eye and camera, bird flying and aircraft flying. 2+4+4=10
- 3) A) Discuss classification of living organisms based on a) Cellularity b) Ultra structure c) Energy and Carbon utilization d) Types of excretion e) Habitat. 2+2+2+2+2=10

Or

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B) What do you mean by bacterial growth kinetics? Explain ecological aspects of single celled organisms. $4+6=10$

4) A) What is Mendelian genetics? What are the different laws given by Mendel? $4+6=10$

Or

B) What do you mean by Cell division in eukaryotes and what are its various phases? Explain Mitosis and Meiosis with suitable diagram. What are the major differences between the two? $3+4+3=10$

5) A) What is hierarchy of protein structure? Explain each type with suitable diagram. $2+8=10$

Or

✓ B) What is cellular respiration? What are the various processes involved in it? Mention the brief summary of each type. $2+2+2+2+2=10$

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